Quantum algorithms through composition of graphs

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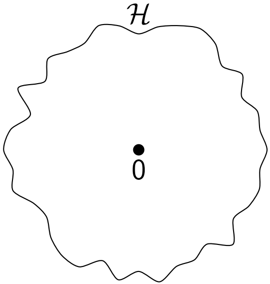

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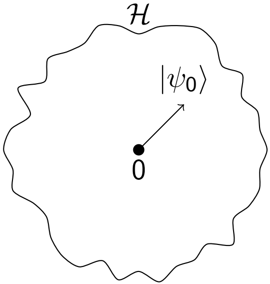

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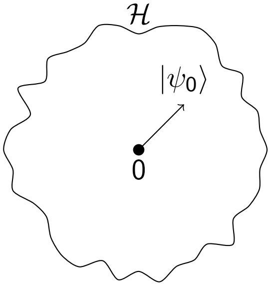

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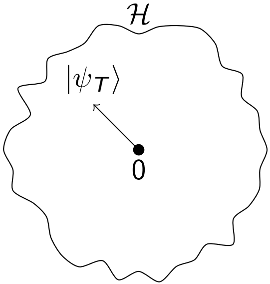

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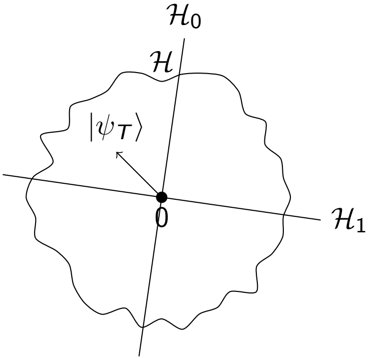

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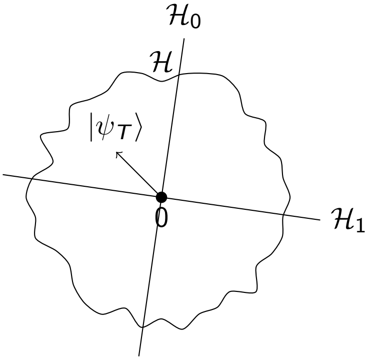

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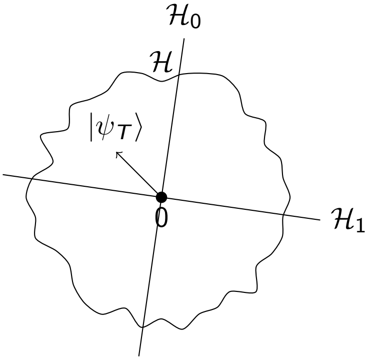

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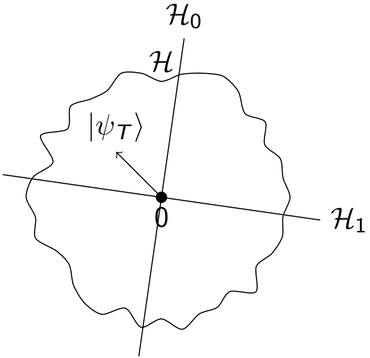
Quantum query complexity: $\mathrm{Q}(f; O_x)$:

1. Minimum number of oracle calls.



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Quantum algorithmic frameworks (for boolean functions)

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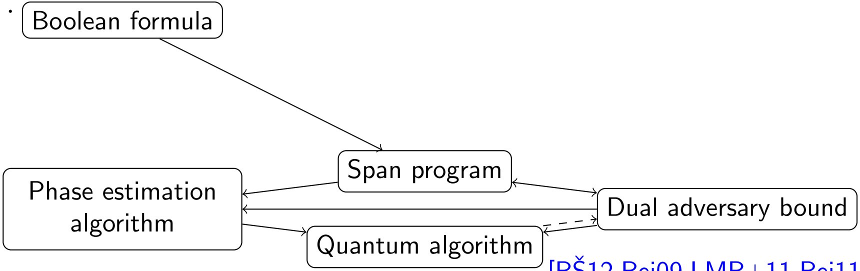

[RŠ12,Rei09,LMR+11,Rei11]

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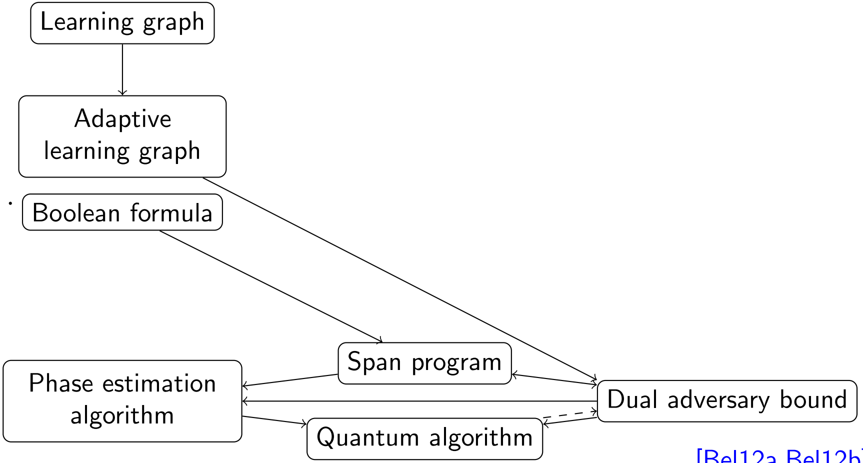

[Bel12a,Bel12b]

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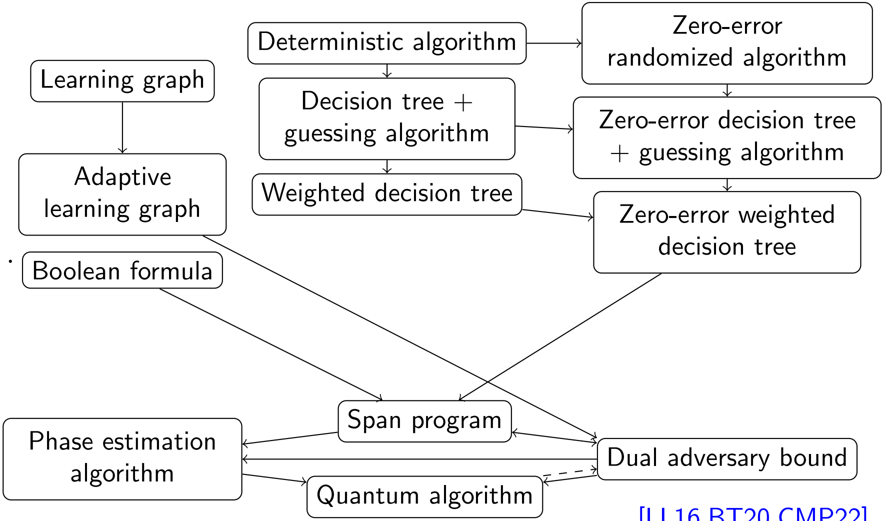

[LL16,BT20,CMP22]

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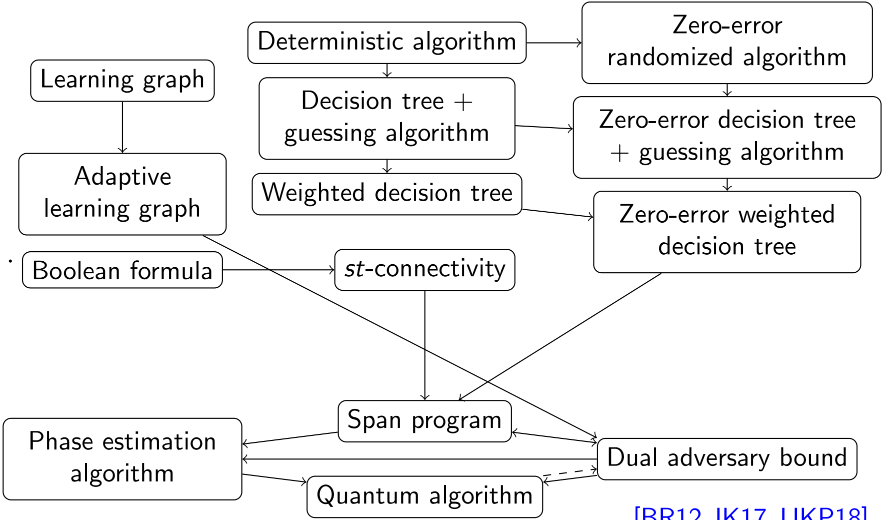

[BR12,JK17,JJKP18]

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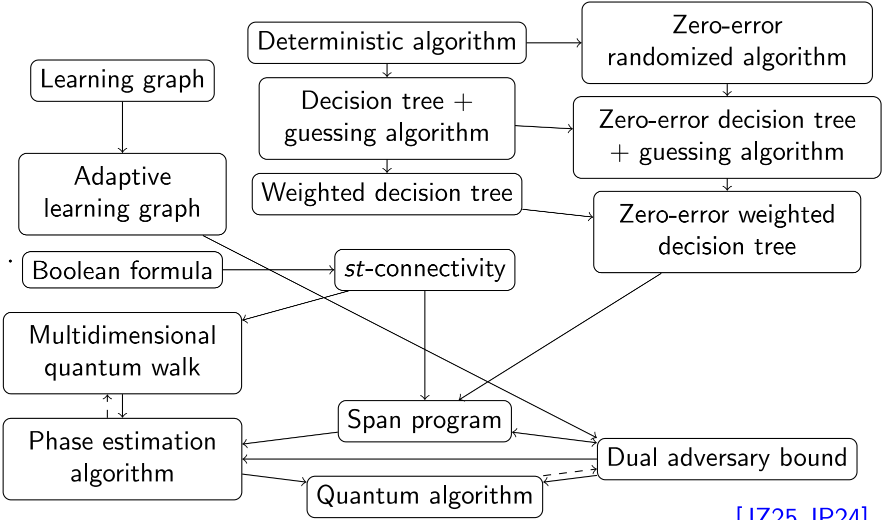

[JZ25,JP24]

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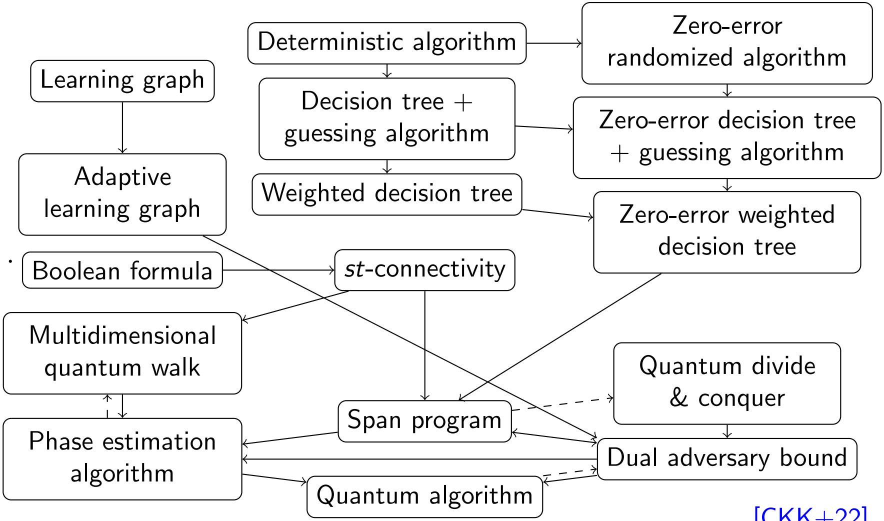

[CKK+22]

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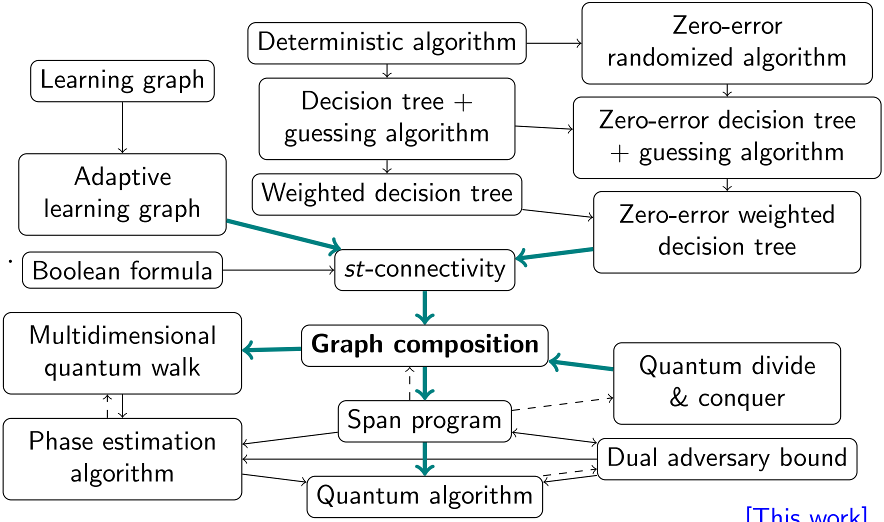

[This work]

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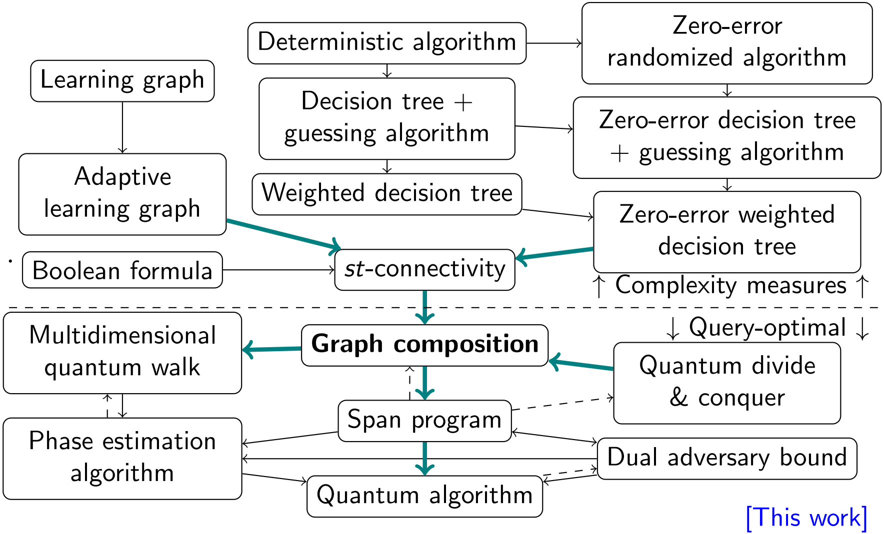

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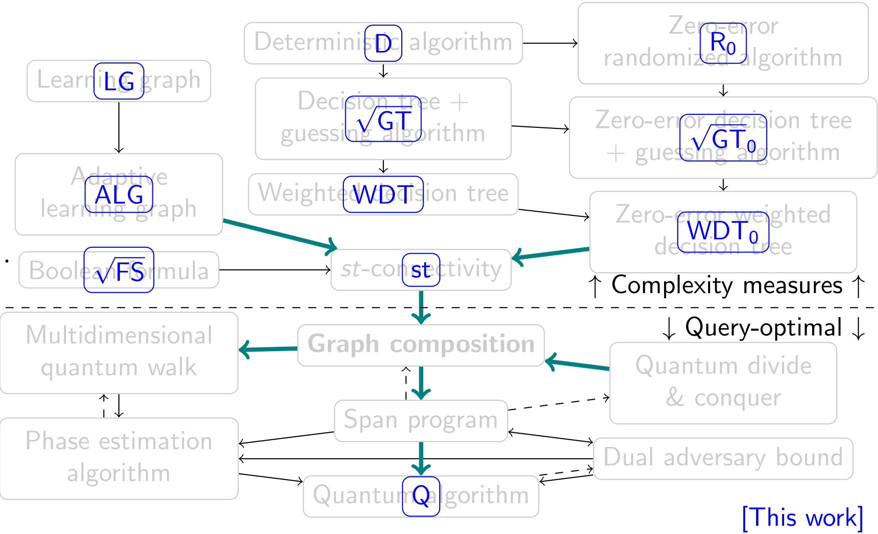
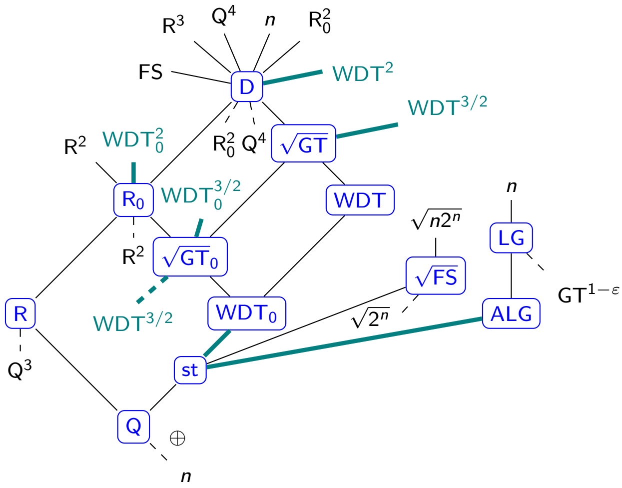


Complexity measure relations for total boolean functions



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A — B: $\forall f : \{0,1\}^n \to \{0,1\}$, $A(f) \in \widetilde{O}(B(f))$

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A ━ B: New in this work

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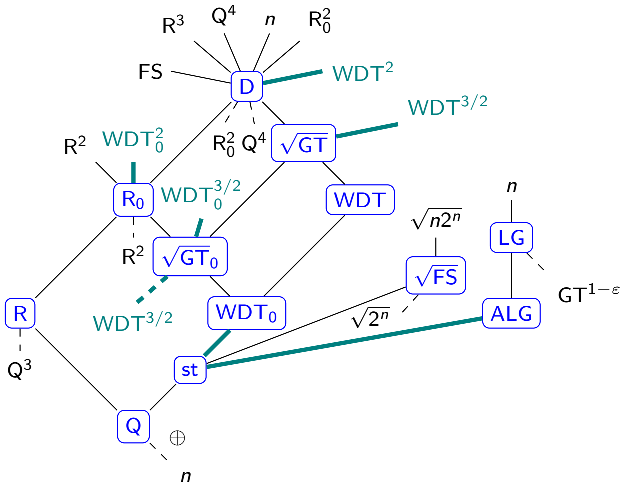

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Open questions:

1. Separation between Q and st?
2. Can we prove $R \in \widetilde{O}(\mathrm{st}^2)$?

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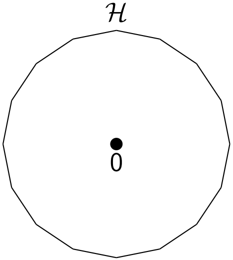

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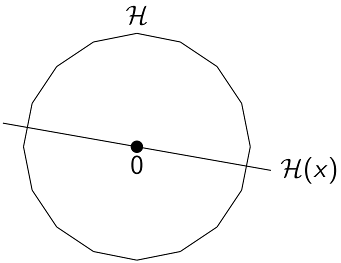

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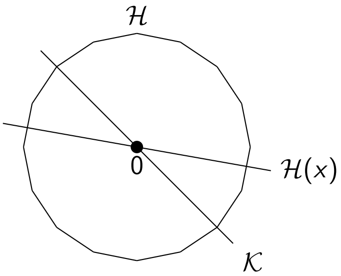

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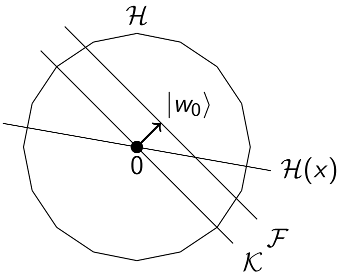

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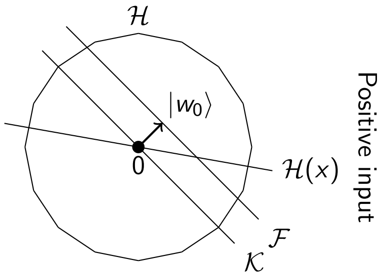

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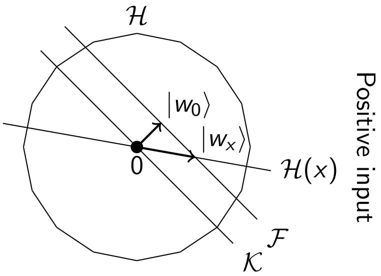

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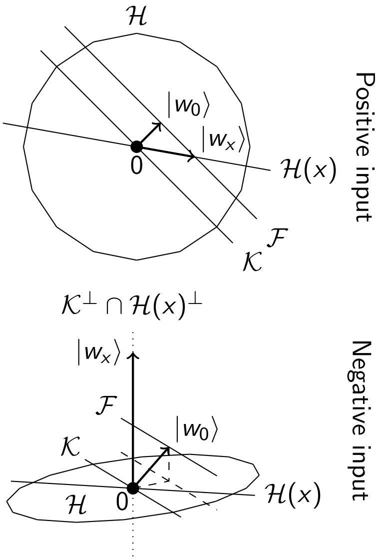

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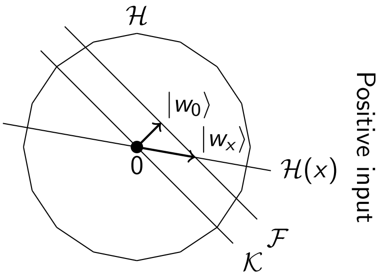

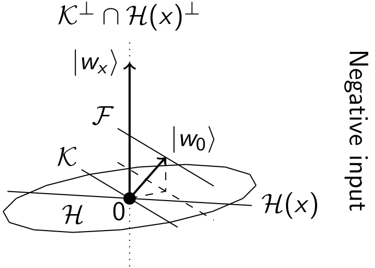

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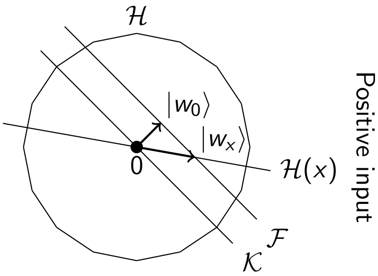
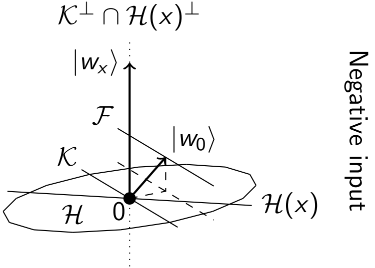
3. $w_-(x,\mathcal{P}) = \min\{\||w\rangle\|^2 : |w\rangle \in \mathcal{K}^\perp \cap \mathcal{H}(x)^\perp, \langle w_0|w\rangle = 1\}$.
4. $C(\mathcal{P}) = \sqrt{\max_{x \in f^{-1}(0)} w_-(x,\mathcal{P}) \cdot \max_{x \in f^{-1}(1)} w_+(x,\mathcal{P})}$.

Thm: $Q(f; 2\Pi_{\mathcal{H}(x)} - I) = O(C(\mathcal{P}))$ [Rei11].





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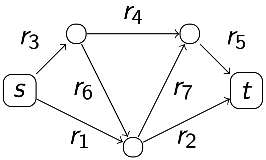
$G =$ 

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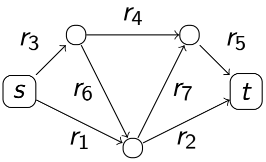
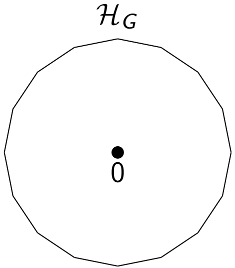
$G =$ 



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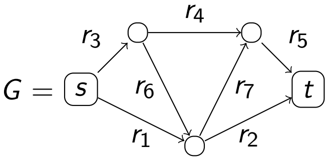

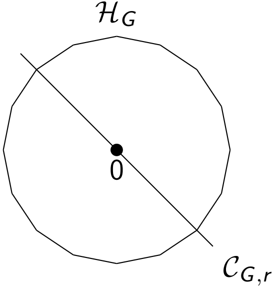

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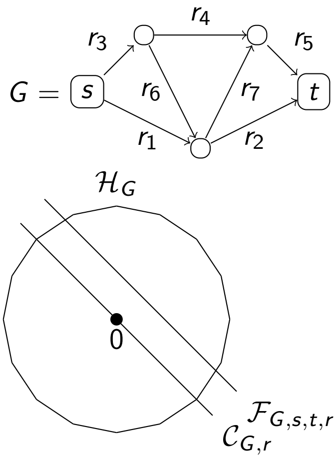

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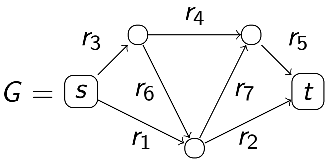

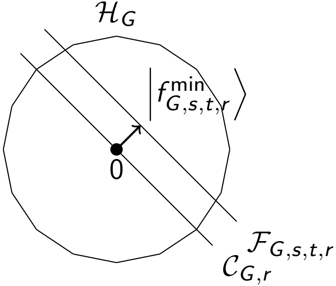

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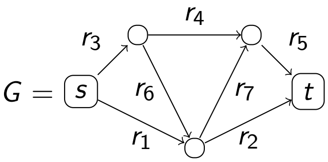

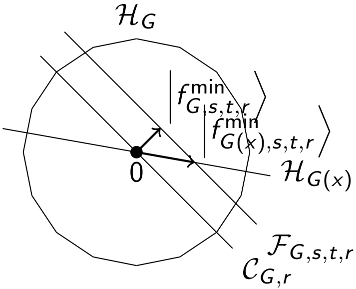

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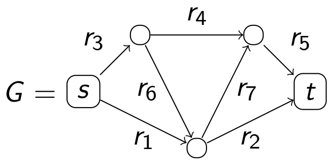

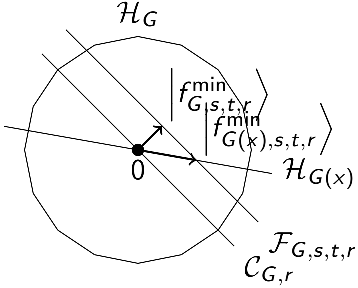

st-connectivity span program: $(\mathcal{H}_G,x\mapsto\mathcal{H}_{G(x)},\mathcal{C}_{G,r},|f^{\min}_{G,s,t,r}\rangle)$.

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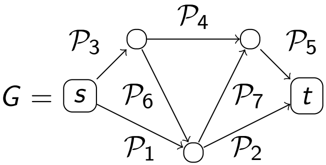

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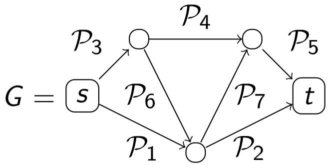

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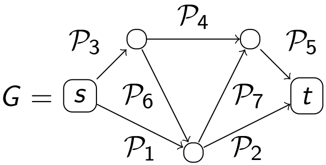

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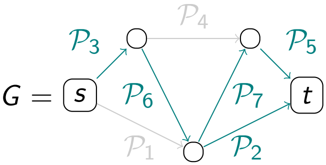

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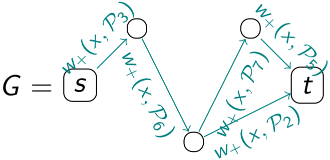

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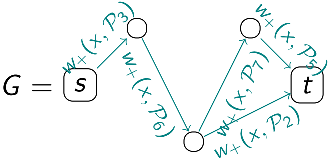

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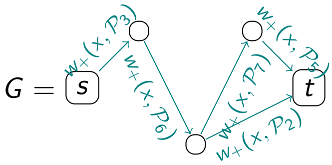

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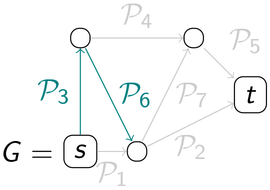

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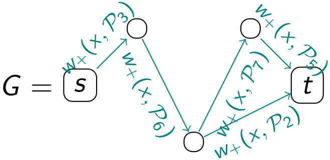

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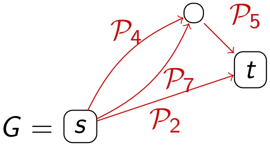

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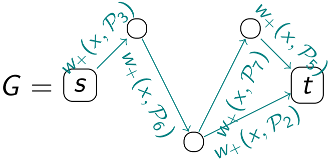

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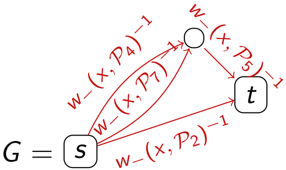

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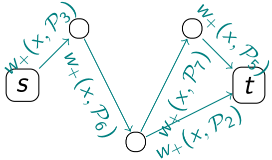
Formally: Span program $\mathcal{P}$ on $\mathcal{D}$:

1. $\mathcal{H} = \bigoplus_{e \in E} \mathcal{H}_e$
2. $\mathcal{H}(x) = \bigoplus_{e \in E} \mathcal{H}_e(x)$
3. $\mathcal{E} : \mathcal{H}_G \to \mathcal{H}, |e\rangle \mapsto |w_0^e\rangle \,/\, \||w_0^e\rangle\|$.
4. $\mathcal{K} = \mathcal{E}(\mathcal{C}_{G,r}) \oplus \bigoplus_{e \in E} \mathcal{K}_e$, with $r_e = \||w_0^e\rangle\|^2$.
5. $|w_0\rangle = \mathcal{E}(|f^{\min}_{G,s,t,r}\rangle)$.

Main theorem: For all $x \in \mathcal{D}$,

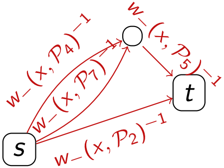
1. $w_+(x, \mathcal{P}) = R_{G,s,t,r^+}$ with $r_+(e) = w_+(x, \mathcal{P}_e)$.
2. $w_-(x, \mathcal{P}) = R^{-1}_{G,s,t,r^-}$ with $r^-(e) = w_-(x, \mathcal{P}_e)^{-1}$.

Positive witness size:

$G =$ 

$$w_+(x, \mathcal{P}) = R_{G,s,t,r^+}.$$

Negative witness size $w_-(x, \mathcal{P})$:

$G =$ 

$$w_-(x, \mathcal{P}) = R^{-1}_{G,s,t,r^-}.$$

Path-cut theorem

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Theorem: For all $x \in \mathcal{D}$,

1. Let P be a path from s to t:
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Path-cut theorem

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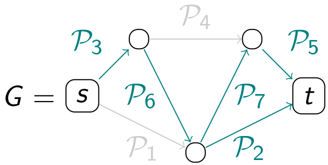

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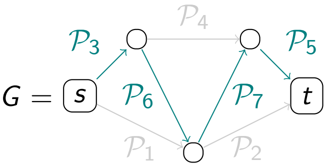

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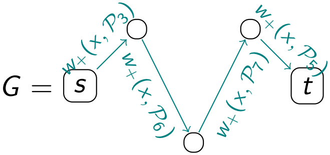

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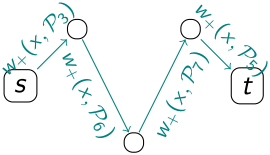
$G =$ 

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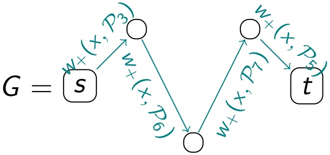

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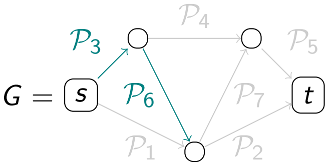
Negative input:



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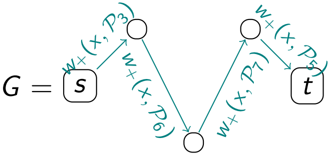

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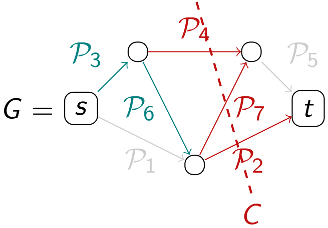

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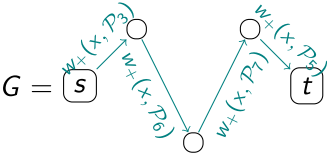

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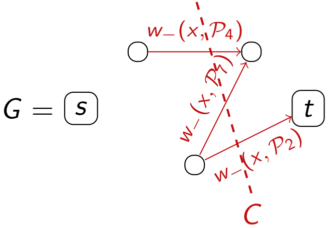

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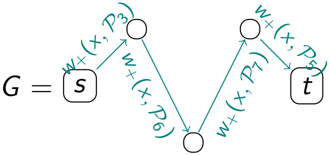

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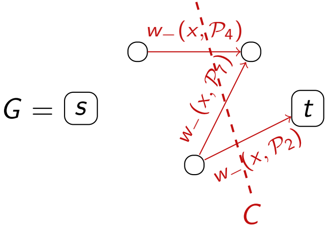

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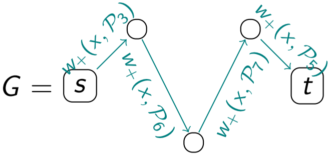
Properties:

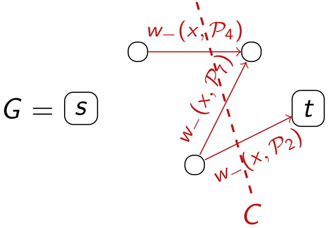
1. Simpler (less-powerful) version.
2. Still powerful enough for many applications.

Positive input:



$$w_+(x,\mathcal{P}) \leq \sum_{e \in P} w_+(x,\mathcal{P}_e).$$

Negative input:



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Example: OR-function

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Trivial span program: $(\alpha > 0)$

1. $x_j : \{0,1\}^n \to \{0,1\}$, $x \mapsto x_j$.
2. $w_+(x, x_j) = \alpha$, if $x_j = 1$.
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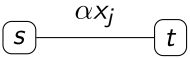

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1. $\mathrm{OR}_n : \{0,1\}^n \to \{0,1\}$
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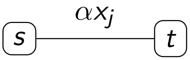

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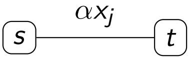

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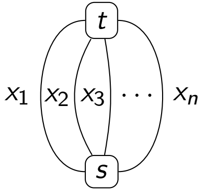
Graph composition for OR_n:



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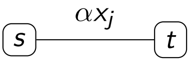

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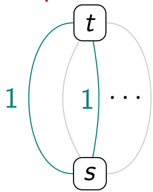

$x = 1010\cdots0 \Rightarrow w_+(x) = \frac{1}{2}$

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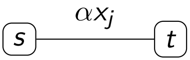

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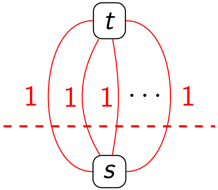

$x = 0000 \cdots 0 \Rightarrow w_-(x) = n$

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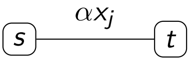
Trivial span program for x_j:



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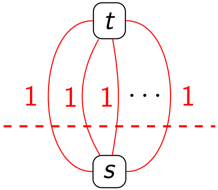

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5. $\Rightarrow Q(\mathrm{OR}_n) \in O(\sqrt{n})$.

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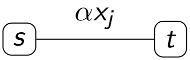

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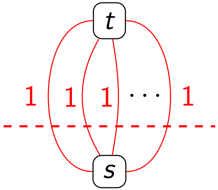

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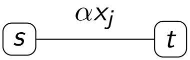
Quadratic speed-up for search.

Trivial span program for x_j:



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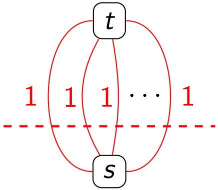

$x = 0000\cdots0 \Rightarrow w_-(x) = n$

Example: Threshold function

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The threshold function: $(k \in [n])$

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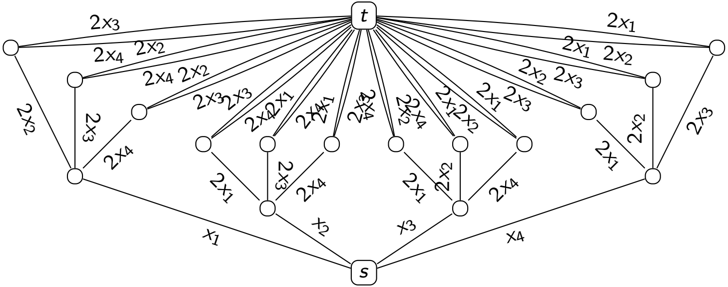
Graph composition for Th_4^3*:*



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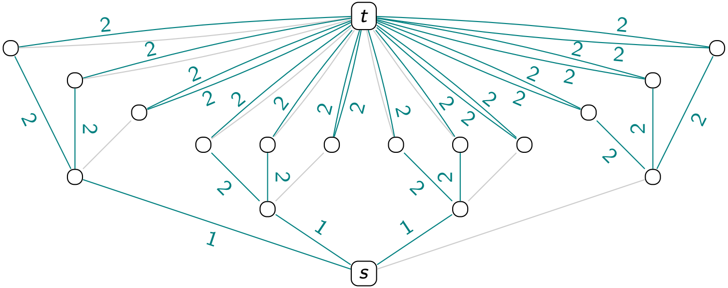
Graph composition for Th_4^3*:*



$x = 1110 \Rightarrow w_+(x) = 1$

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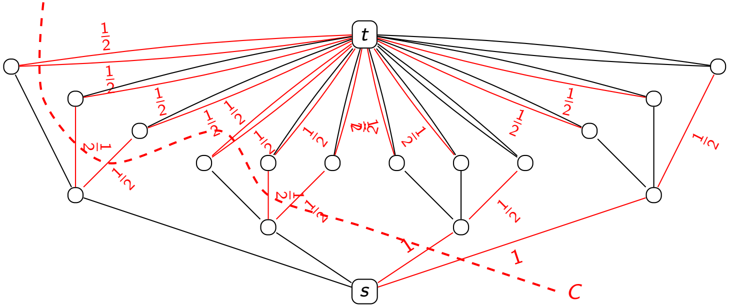
3. $w_-(x) = \frac{k(n-k+1)}{k-|x|}$

Graph composition for Th_4^3*:*



$x = 1100 \Rightarrow w_-(x) = 6$

Example: Threshold function

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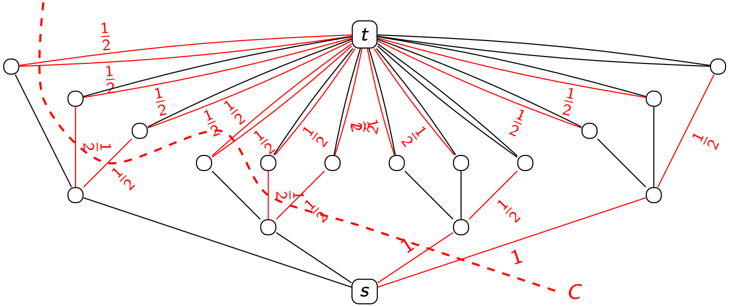
Graph composition for Th_4^3*:*



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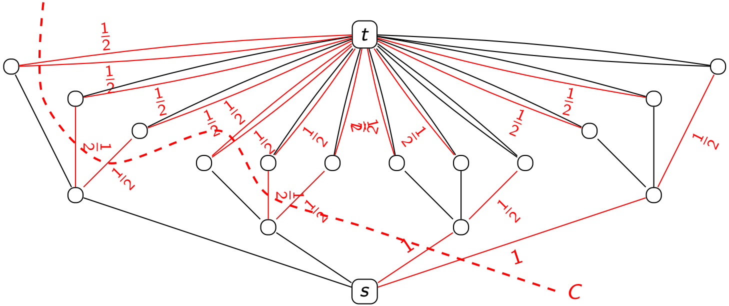

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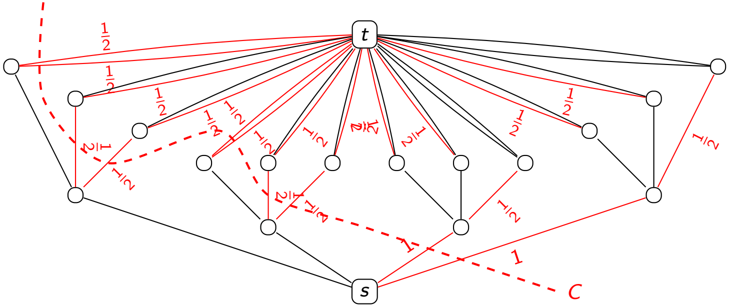

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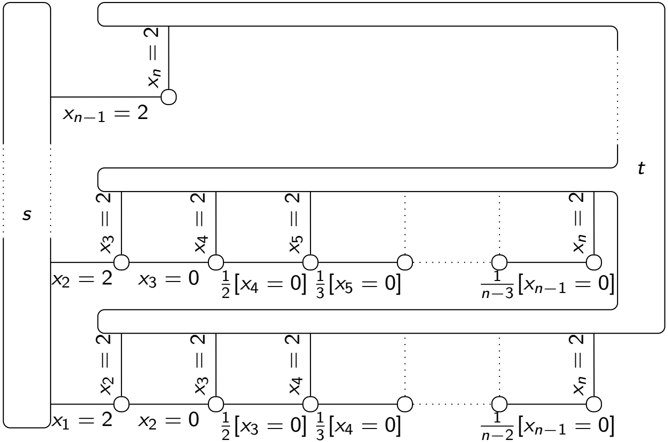

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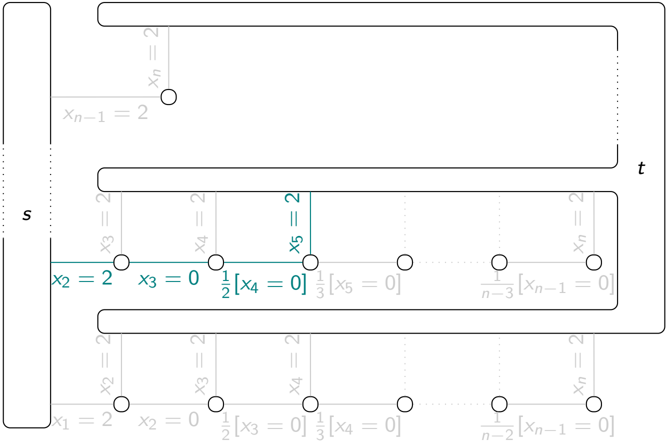

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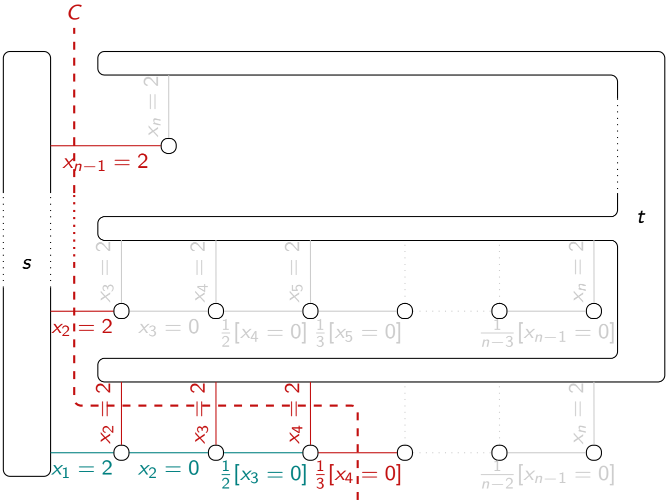

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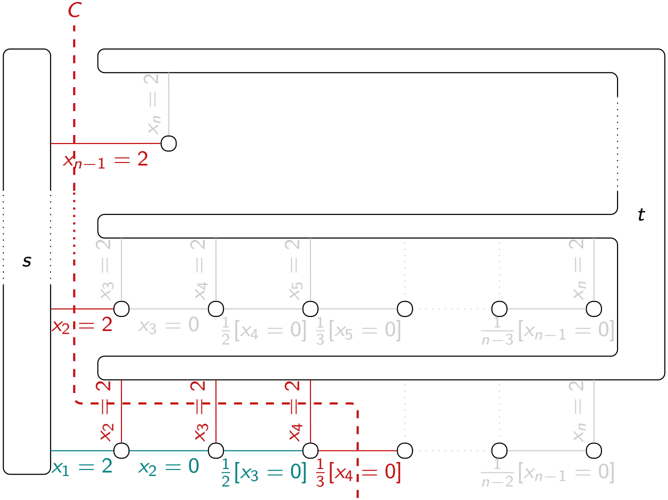

Summary (I/II)

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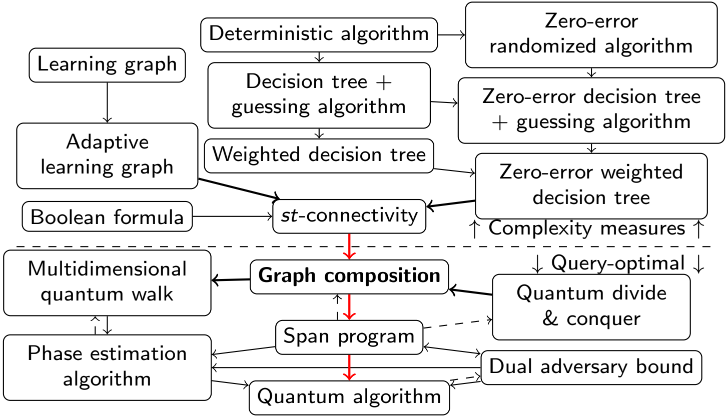

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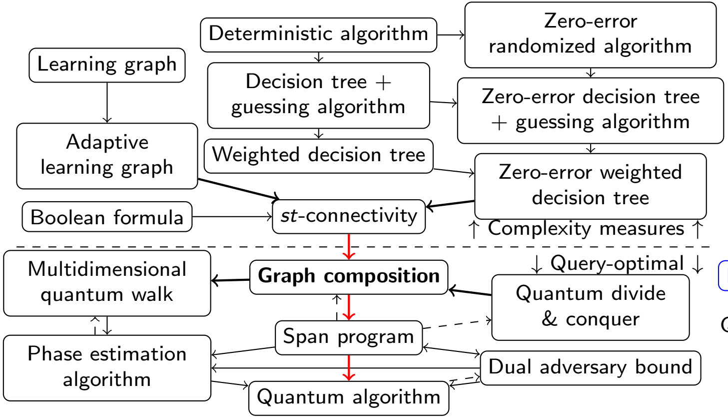
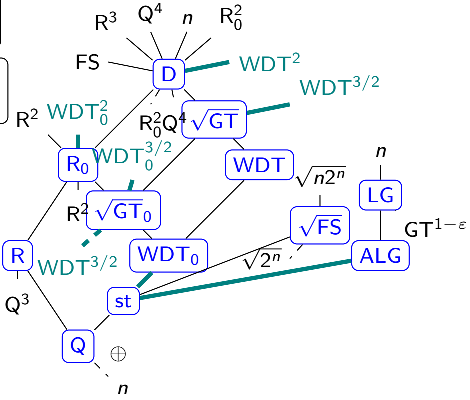


Relations between complexity measures



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Thanks for your attention!

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References (I/III)

[AGJ21] Simon Apers, András Gilyén, and Stacey Jeffery. A unified framework of quantum walk search.

[Bel12b] Aleksandrs Belovs. Span programs for functions with constant-sized 1-certificates.

[Bel12a] Aleksandrs Belovs. Learning-graph-based quantum algorithm for k-distinctness.

[BR12] Aleksandrs Belovs and Ben W Reichardt. Span programs and quantum algorithms for *st*-connectivity and claw detection.

[BT20] Salman Beigi and Leila Taghavi. Quantum speedup based on classical decision trees.

[CKK+22] Andrew M Childs, Robin Kothari, Matt Kovacs-Deak, Aarthi Sundaram, and Daochen Wang. Quantum divide and conquer.

[CMP22] Arjan Cornelissen, Nikhil S Mande, and Subhasree Patro. Improved quantum query upper bounds based on classical decision trees.

[JJKP18] Michael Jarret, Stacey Jeffery, Shelby Kimmel, and Alvaro Piedrafita. Quantum algorithms for connectivity and related problems.

References (II/III)

[JK17] Stacey Jeffery and Shelby Kimmel. Quantum algorithms for graph connectivity and formula evaluation.

[JP24] Stacey Jeffery and Galina Pass. Multidimensional quantum walks, recursion, and quantum divide & conquer.

[JZ25] Stacey Jeffery and Sebastian Zur. Multidimensional quantum walks, with application to k-distinctness.

[LL16] Cedric Yen-Yu Lin and Han-Hsuan Lin. Upper bounds on quantum query complexity inspired by the elitzur–vaidman bomb tester.

[LMR+11] Troy Lee, Rajat Mittal, Ben W Reichardt, Robert Špalek, and Mario Szegedy. Quantum query complexity of state conversion.

[Rei09] Ben W Reichardt. Span programs and quantum query complexity: The general adversary bound is nearly tight for every boolean function.

[Rei11] Ben W Reichardt. Reflections for quantum query algorithms.

References (III/III)

[RŠ12] Ben Reichardt and Robert Špalek. Span-program-based quantum algorithm for evaluating formulas.